Mahmoud Alyosify

AI & Machine Learning Engineer | LLM Systems, Applied ML & Model Deployment
MSc Artificial Intelligence, Queen's University

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PROFESSIONAL SUMMARY

AI & Machine Learning Engineer who builds intelligent systems end to end — from research question to a deployed, measured model. Currently an **AI Engineer Intern at Injaz Tech**, building agentic RAG systems with **LangChain, LangGraph and FastAPI**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**; master's research (**MinimalLM**) tests training-free decoding-time control of LLM verbosity against a strict truncation baseline. Core strengths in **LLM fine-tuning, RAG and inference optimisation**, **end-to-end ML pipelines** and **cloud deployment on AWS**, backed by rigorous, reproducible evaluation. Prior engineering experience on national-scale toll infrastructure, plus six years teaching AI and computer science to **10,000+ learners** (**700K+ views**), including part-time instruction at NTI. Open to AI/ML roles in Egypt, the Gulf, remote or international.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation from video via rPPG. **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

AI & LLM Systems	Large Language Models, Fine-Tuning (LoRA, QLoRA, PEFT), RAG, LangChain, LangGraph, Transformers, Hugging Face, Ollama, Hybrid Search & Cross-Encoder Re-Ranking, Prompt Engineering, Structured Outputs, Multi-Agent Systems, Agentic Workflows
ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, Computer Vision, NLP, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Optuna, SHAP
Data & MLOps	Pandas, NumPy, PySpark, ETL Pipelines, Feature Engineering, AWS, Docker, CI/CD (GitHub Actions), pytest, FastAPI, ONNX, Streamlit, FAISS, ChromaDB, Prometheus, MySQL, MongoDB, Git/GitHub, Linux
Evaluation	Benchmarking (MT-Bench, XSum, SQuAD-v2), BERTScore, ROUGE, LLM-as-Judge, RAG Evaluation (Groundedness, Top-1 Accuracy, MRR), Statistical Testing, Ablation Studies, Reproducible Experiment Design
Programming	Python, SQL, C++, C, C#, Go, JavaScript, MATLAB, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built an **agentic RAG assistant** for internal-docs Q&A (**LangChain**, **LangGraph**): header-aware chunking, **FAISS** search with **cross-encoder re-ranking**, LLM relevance grading, query rewriting, hallucination checks, chat memory and verified citations.
- Turned it into a **FastAPI** service (SSE streaming, caching, retries, circuit breaker, rate limiting, prompt-injection guardrails, Prometheus metrics) with **Docker**, **136 pytest** tests and a **GitHub Actions** evaluation gate.
- Drove design choices with measurement: raised top-1 retrieval accuracy from **75% to 94%**, cut false refusals from **5/20 to 1/20** by redesigning the LLM graders, and cut repeat-question latency from **34.7 s to 1.6 s** with a context-aware answer cache.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Cairo, Arish & Assiut, Egypt

- Teach **Machine Learning** (90-hour bootcamps in Cairo, Arish and Assiut) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
- Co-developed an internal data-management and reporting desktop application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**; supported and troubleshot legacy applications to minimise operational downtime.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built and taught Arabic-language technical curricula (discrete mathematics, algorithms, probability & statistics, graph data mining) reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED PROJECTS

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation *2026 – Present*
MSc Research Project (CISC 898), Queen's University · PyTorch · Hugging Face · NVIDIA A40 GPU

- Research on **training-free decoding-time control** of LLM verbosity, tested against plain truncation on **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B**. An entropy-gated **logits processor** cut **25.5–34.9%** of MT-Bench tokens within pre-registered BERTScore margins, but an audit showed the saving was truncation.
- A survey of **23** training-free mechanism families found none that beat truncation without oracle labels. The proposed **Entailment-Licensed In-Decoding Elision** (sentence encoder + NLI model) beats truncation on every model, removing 0.9–2.2% of tokens; scoping it to discourse units cut reader-judged harmful elisions from 49–69% to **10.2%**.

HORUS Sentinel — Autonomous Multi-Agent OSINT & Threat-Intelligence Platform *2026*
Python · Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected an autonomous intelligence analyst: a **swarm of specialised agents** continuously collects open-source intelligence and correlates findings into a **living knowledge graph**, over which a self-hosted fine-tuned LLM produces **prioritised, evidence-backed reports**.

SimCLR-Vision-SSL — Self-Supervised Contrastive Representation Learning *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Built a self-supervised vision pipeline with a **42-experiment augmentation sweep**, a semi-supervised **SupCon** extension and deployment-ready **ONNX/FAISS** retrieval, reaching **84.30% top-1 accuracy**.

RL-Cloud-Autoscaler — Cloud Resource Provisioning via Deep Reinforcement Learning *2026*
Python · PPO · DQN · Gymnasium · Cost Optimisation

- Built a **custom Gymnasium environment** and trained **PPO** and **DQN** agents to optimise the **cost–latency trade-off** under fluctuating demand, benchmarked against static and threshold-based autoscaling baselines.

Additional Projects

- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records and deployed it on **AWS** as an interactive threat-intelligence assistant.
- **Automated Multimodal Agent** — agentic workflow converting PDFs into narrated PowerPoint decks end to end.
- **SCRAP** — supervised ML pipeline predicting satellite collision risk **48+ hours** before closest approach.
- **AudioShield** — CNN-based deepfake audio detection served through a Streamlit application.
- **Also:** Human vs. Agentic Pull Requests (GitHub mining) · Vitalism Solution (rPPG) · Diabetes Onset Prediction · ATM Transaction Analysis · E-Commerce Web App · Mini-FAT File System.

CERTIFICATIONS

AWS Academy — ML Foundations, ML for NLP, Data Engineering
HCIP – AI (Huawei)
AI in Cybersecurity — NTI (420 hrs)
Generative AI & Prompt Engineering — IBM / Coursera
Machine Learning Specialization — DeepLearning.AI
Microsoft Certified: Data Analyst Associate
ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
Full Stack Diploma — .NET & Angular (Route Academy)

ACHIEVEMENTS & AWARDS

1st Place — Smart Cities Hackathon 2022
Finalist — ISEIC 2023
Semi-Finalist — Microsoft Imagine Cup 2023
Ideal Student Award — Assiut University, 2022–2023

LEADERSHIP & LANGUAGES

- **Secretary**, Higher Scientific & Technological Committee, Student Union, Assiut University · **Ambassador**, Information Technology Institute (ITI), Assiut · **Volunteer Team Leader**, USAID University Center for Career Development.
- **Languages:** Arabic (Native) · English (Professional Working Proficiency) · French (Basic).